

List of Publications

- [1] E.D. Sontag. *Notes on Mathematical Systems Biology (online)*. Online only., 2026. Continuously updated. If the link does not work, then copy/paste this: <http://drive.google.com/drive/folders/1lIRqaCPeXmVZGoY-44bBsvtnsHtlRfIO?usp=sharing>.
- [2] A.C.B de Oliveira, M.K. Wafi, and Eduardo D. Sontag. On input-output persistency and the interconnection of positive nonlinear systems. *arXiv*, page 2608.01699, 2026.
- [3] Eduardo D. Sontag. Smooth globally pli functions are nonlinear least-squares, and so are their gradient-dominated cousins. *arXiv*, page 2608.08849, 2026.
- [4] Eduardo D. Sontag. Some intuition for why cooperative systems "look 1-dimensional" and 2-cooperative systems "look 2-dimensional". *arXiv*, page 2607.27176, 2026.
- [5] E. D. Sontag. Dynamics and dose response in scaffold ligand binding. 2026. Submitted. Preprint in arXiv:2508.06599.
- [6] K. Manoj, D. Jatkar, M. Ali Al-Radhawi, E.D. Sontag, and D. Del Vecchio. Paradoxical gene regulation explained by competition for genomic sites. 2026. Submitted. Also bioRxiv 2025.11.27.691022.
- [7] M.K. Wafi, A.C.B de Oliveira, and E.D. Sontag. Boundedness of solutions in feedback systems with antithetic controllers. 2026. Submitted. Preprint in arXiv 2604.27290.
- [8] M.K. Wafi, A.C.B de Oliveira, and E.D. Sontag. When is cumulative dose response monotonic? analysis of incoherent feedforward motifs. In *Proc. 65th IEEE Conference on Decision and Control (CDC)*, 2026. To appear. Also arXiv:2604.01573.
- [9] A. Oliveira, A. C. B. de Oliveira, M. Sznaier, and E. D. Sontag. On incremental and semi-global exponential stability of gradient flows satisfying generalized lojasiewicz inequalities. In *Proc. 65th IEEE Conference on Decision and Control (CDC)*, 2026. To appear. Also arXiv arXiv:2603.25822.
- [10] A. C. B. de Oliveira, R. Wang, I.R. Manchester, and E. D. Sontag. Remarks on Lipschitz-minimal interpolation: Generalization bounds and neural network implementation. In *Proc. 65th IEEE Conference on Decision and Control (CDC)*, 2026. To appear. Also arXiv:2603.19524.
- [11] E.D. Sontag. Dynamic response phenotypes and model discrimination in systems and synthetic biology. 2025. Submitted. Preprint in arXiv 2512.24945.
- [12] D.D. Jatkar, K. M. Aravind, E. D. Sontag, and D. Del Vecchio. Paradoxical gene regulation explained by competition for genomic sites. *bioRxiv, being submitted.*, page 2025.11.27.691022, 2025.
- [13] A.P. Tran, D.D. Jatkar, M.A. Al-Radhawi, E. Ernst, and E.D. Sontag. Optimization of heuristic logic synthesis by iteratively reducing circuit substructures using a database of optimal implementations. *bioRxiv*, page 2025.11.28.691216, 2025.
- [14] A.C.B de Oliveira, D.D. Jatkar, and E.D. Sontag. On the convergence of overparameterized problems: Inherent properties of the compositional structure of neural networks. In G Sukhatme, L Lindemann, S Tu, A Wierman, and N Atanasov, editors, *Proceedings of The 8th Annual Learning for Dynamics and Control Conference*, volume 331 of *Proceedings of Machine Learning Research*, pages 1088–1107. PMLR, 17–19 Jun 2026. Also 2025 arXiv:2511.09810 [cs.LG].
- [15] M. D. Kvalheim and E. D. Sontag. Autoencoding dynamics: Topological limitations and capabilities. *arXiv*, page 2511.04807, 2025. Journal version submitted.
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- [18] L. Cui, Z.P. Jiang, and E. D. Sontag. Small-covariance noise-to-state stability of stochastic systems and its applications to stochastic gradient dynamics. In *2026 American Control Conference (ACC)*, pages 2825–2830, 2026. Also 2025 arXiv:2509.24277.
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- [20] E. D. Sontag. Dynamics of binding three independent ligands to a single scaffold. *arXiv*, page 2508.06599, 2025.
- [21] M.K. Wafi, A.C.B de Oliveira, and E.D. Sontag. On the (almost) global exponential convergence of overparameterized policy optimization for the LQR problem. In *2026 American Control Conference (ACC)*, pages 2819–2824, 2026. To appear. See also 2025 arXiv:2510.02140.
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